

FossilJet: AI-Driven Farm Optimization

[0:00] And so we then produce this intelligence and share this intelligence with farmers on their mobile phones, on their WhatsApp, wherever they are, but in very simple manner that they can consume.

[0:13] Somebody who doesn't even know how to read can just hear.

[0:17] And then the next thing is how do we build machines and technologies which are relevant for India, can survive Indian margin pressure, can build an Indian business but execute all these decisions without the farmer having to have the worry of whether there is labor to execute that task or not.

[0:38] The first thing that we have done is execute irrigation and execute fertigation, which is putting water without any labor and putting water fertilizer without any labor.

[0:50] We have deployed about 14,000 IoT devices, this would also be the biggest IoT deployment at least in this part of the world for farm.

[1:00] These devices collect a lot of data about canopy's temperature, the leaf wetness on the leaves, the moisture in the soil in real time every hour, every half an hour.

[1:10] We have about 10 billion data points like that now.

[1:13] We've deployed it across 200,000 acres of farmland with individual farmers plus 120 enterprises.

[1:21] We have been collecting this data for last five, seven years.

[1:24] It has become very big data sets also has created a lot of different issues.

[1:28] And then act is where FossilJet and FossilJet Pro come into picture.

[1:32] They can act on the recommendation that system is generating.

[1:38] This is how the machine that does irrigation and fertigation look like.

[1:41] And the result of the technology in the last three, four years is that we have

saved about 83 billion liters of water.

[1:47] So this is the water that ideally you would get to drink but goes in the farms.

[1:53] 83 billion liters of water if you want to imagine, imagine the Powai lake and multiply it by 83 times.

[1:59] About 210,000 kilograms of pesticides that you could have ultimately consumed was avoided on these farms.

[2:07] 56,000 metric tons of GHG emissions, so think of entire Mumbai or think of Kolhapur as an area and assume that the air was clean, so we consumed, sucked up everything that was making the air not so clean.

[2:20] And this is a multi-country approach.

[2:22] I mean, the fact is what you build in India is very relevant for a lot of parts of the world.

[2:31] You can hear directly from what the technology does to a very average farmer who probably doesn't even know how to speak anything beyond his local language.

[2:42] Perhaps you can play this now.

[2:53] Maro naam Kishor Kumar Tulsi Das lakhani gaon pantiya sarvath mein me jare kheti karta jare ye normally sada pakni jo avtar karta bejar no aaye pachme daadavni kheti chalu karisho me daadavni kheti ma jo hai toh sarvath ni andar hamare rog jeevan pochha avta atyadina samay ma jo hai natyadina climate praman jo hai toh rog jeevan atyadina toh udar rehte ho.

[3:14] Translators note.

[3:44] It's.

[4:14] It.

[4:43] Uttar Pradesh.

[5:14] And.

[5:27] He said, I'm going to go to the village.

[5:35] I'm going to go to the village.

[5:42] I'm going to go.

[5:56] There are about 13,000 of them right now.

[5:59] We are hopeful that by end of this financial year, we should be about 20,000.

[6:04] And I see the next 10 years for us as well as the world, a journey of taking all of this screen intelligence to something that truly can work on the farms in complex situations as well.

[6:19] This is us.

[6:21] we are a team of about 120 odd people, patented technology working with farmers since 2022 in about 30 odd districts in India and about six countries now.

[6:32] That's all for me.

[6:33] Thank you so much.

[7:01] Thanks for the good session, Shailendra.

[7:05] I think first if you could just let us know what the business model is.

[7:07] So how do you charge the farmers?

[7:09] Is it for the device and then incrementally for data sharing and reports and analysis with it?

[7:14] No, certainly.

[7:15] So we sell the hardware.

[7:17] We also attempted subscriptions, but we realized subscriptions are very, very

difficult.

[7:24] And farmers are technically more comfortable with CAPEX than OPEX.

[7:28] So right now we are on our journey of converting that OPEX to CAPEX.

[7:34] So we sell the hardware, all kinds of hardware, and then there is a small percentage of subscription over the life of the hardware.

[7:45] And does this model, as you said, you focus on the horticulture trade?

[7:48] Is it just primarily, as you said, because much more value per acre is there and hence it makes commercial sense to introduce it there?

[7:56] Or do you see this as sensor costs come down, as you said, can this be used for wheat, rice, etc.?

[8:03] See, the technology can be applied on any crop because the fundamentals remain the same.

[8:08] Now there are two things.

[8:09] One is who is ready to adopt technology?

[8:13] Somebody who's yield grows by 20% and makes 2.5 lakh rupees or 1 lakh rupees additional per acre has an inherent incentive of looking for these technologies and adopting them and scaling them.

[8:28] The challenge that I see with crops like wheat, rice, etc., at least directly with the farmer, so within agricultural ecosystem, there are variety of customers.

[8:37] You have the farmer, you have the district, you have the government.

[8:41] And everybody has a lot of vested interest, so I'll share something very interesting.

[8:46] The government of India's agriculture budget is 1,22,000 crores.

[8:51] Additionally, the fertilizer subsidy budget is 1,70,000 crores.

[8:56] So the budget for fertilizer subsidy is bigger than the agri budget for an entire year.

[9:02] And so from that point of view, the government could pay.

[9:06] But from a farmer's standpoint, what I still continue to struggle with is the cost of explaining the value of the technology in itself is a deal breaker right now.

[9:17] Maybe five years, seven years down the line, there would be a different business model which would enable something.

[9:26] The technology is applicable in all crops.

[9:29] The question is who is ready to adopt and where should I focus very limited resources that I have.

[9:38] And you quantify the impact it had on pesticide usage, water usage, possible to quantify the impact it had on yield, let's say, and fertilizer usage.

[9:48] Certainly.

[9:48] So in general, we see a 15% to 300% improvement.

[9:57] And it depends on crop to crop.

[9:58] So 300% improvement not because we are doing something miraculous, but the way in which we grow crops is very bad.

[10:05] Let's say in vegetables, I have seen on an average about 20, 25% improvement.

[10:11] In fruits, I've seen anything between 10 to 20% improvement, which is enough and it recovers the cost of the technology for all of these crops within a year.

[10:21] In terms of fertilizers, We don't directly measure because we can't measure.

[10:27] For water, we can directly measure for GHG emissions, this is the number of

space, sprays reduced, number of pump outs reduced, we can directly measure.

[10:36] There is an estimation that we have saved about 15% fertilizer every time on every farm, but that's a non-audited data.

[10:49] The numbers that I'm sharing are audited numbers.

[10:53] Another audio question I see is, as you mentioned, you have lots of sensors spread across the country now, and you're collecting tons of data on this.

[11:04] Can that data be used for other reasons or be sold commercially?

[11:08] Absolutely, yes.

[11:09] And I think that's the bigger, more beautiful part of this entire play.

[11:15] What we are collecting is a lot of climate, farm, and crop data.

[11:21] Climate farm and crop data can be used in variety of ways.

[11:24] The easiest, so for example, some of you might be aware that IMD has about 4,000 weather stations in the country.

[11:30] They have been installing weather stations from 1947, I believe.

[11:35] We have about 14,000 now.

[11:39] So the country needs at least 200,000 stations before we can make any reasonably good weather prediction.

[11:45] The reason why we don't make very good weather prediction is because we don't have the infra.

[11:50] So that's point number one.

[11:52] Second is a lot of commercial usage of this data.

[11:54] So for example, at this point in time, I know how many farmers are having a

good season for growing grapes in Nasik.

[12:02] I know whether the prices will be good or bad because you have a representative data of whether the season allowed for a good crop to happen or not.

[12:11] And so that's very useful for buyers.

[12:13] That's very useful for supply chain players who want to proactively manage their procurement or input companies so input companies decide what they are going to place on the stores in retail chains this is what happened last year they lose sales because there are often times where you need very specialized high margin products which is not available because nobody has the real time data so there

[12:37] is a variety of application of this data there is carbon credits water credits data so every

[12:43] every farm we go to have reduced water consumption at least by 30%.

[12:46] That in itself converts to a lot of savings.

[12:49] So that's another angle that we are looking to play and probably that could be a very large angle to play as well.

[12:54] But right now you're not commercializing it.

[12:56] Let's say not right now.

[12:58] So the question was, do you have large enough data set?

[13:01] And I think that question is now answered.

[13:04] Now we're trying to pursue some of those.

[13:06] Perfect.

[13:07] Makes sense.

[13:07] And were you going to customize the solution region to region?

[13:11] Each region in India is different, the crops are different.

[13:13] So it's a myth that you need to customize for countries and regions.

[13:20] It's to say that if I get cold here in India, am I going to get cold in Australia or not?

[13:29] The fundamentals of science across the world within the physical boundaries of biology are same.

[13:36] A plant requires water for the same reason in India as it requires it in Kenya or as it requires it in Europe.

[13:44] That mathematics doesn't change.

[13:46] So what changes is very minor delta on what we call resistance building.

[13:56] So in certain areas you might have very robust plant varieties which are resistant to a certain problem.

[14:05] That's a very easy thing.

[14:06] And you see also the fact that we did R&D for five to four years before we commercially launched.

[14:11] was because we wanted to build science first models which will scale across other parts of the world.

[14:16] So the data that is needed is actually very small.

[14:19] It is not more than 15% of work that needs to be redone.

[14:23] Makes sense.

[14:24] Another question I see here is you have 100 models as you said.

[14:27] How do you decide which model to use and when given a wrong decision can impact crop season?

[14:34] I can go back.

[14:39] Not my slide.

[14:51] I'm going to use a slide to paint a picture of when for what do you need a model.

[14:55] See essentially we are working on the growing phase of the crop where the farmer is growing the crop.

[15:07] The farmer is growing the crop and these are decisions that people have to take every day no matter what.

[15:13] They have to decide whether there is requirement for water.

[15:16] It's a non-negotiable.

[15:16] It's like if you don't feed the crop, it is going to die.

[15:21] Then managing disease, so the crops will get sick.

[15:23] That is the nature of biology.

[15:26] So you have to protect them from disease, pest issues, climate risk issues.

[15:30] So for every crop, and every crop at variety of its crop stages are different.

[15:35] So you might see a small plant like this, it will require very different amount of water.

[15:39] And a tree, 10 feet tall, will require very different amount of water.

[15:44] So what we need to do is number one, predict the stage of the crop without seeing the crop on the farm.

[15:50] So that's algorithm number one that is common across all the crops.

[15:54] Second is irrigation requirement.

[15:55] So by crop, by stage of the crop, by different texture of soil, in different locations on different altitude, you have to figure out how much water the crop needs.

[16:06] So that's the second algorithm.

[16:09] the 200 algorithms that I spoke about are disease and pest algorithms so that's something which is different so you might have some diseases here in India and some additional diseases in Thailand which one we keep or we build is a function of which market we are dealing with and then climate risk management is fundamentally a common subject I mean it's not just India every part of the world where

[16:33] climate was something that farmers intuitively understand now there is

[16:38] 35 degrees centigrade in a place in Thailand where the temperature shouldn't go beyond 24, 25.

[16:44] So the climate risk management has also become a common theme across.

[16:47] And these are the models that we use.

[16:50] Perfect.

[16:51] I think those were the questions that we wanted to cover.

[16:53] Thank you so much.

[16:54] Thank you so much.

[16:55] Thank you everybody for your time.

[17:05] Do you want water?

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